

Visualizing the Impact Of Winning On National Football League Franchise Valuations

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ABSTRACT

With the National Football League generating nearly 23 billion dollars in total revenue, marking a record high for the 2nd consecutive year in a row, the NFL represents the largest and most financially successful sports business in the world.

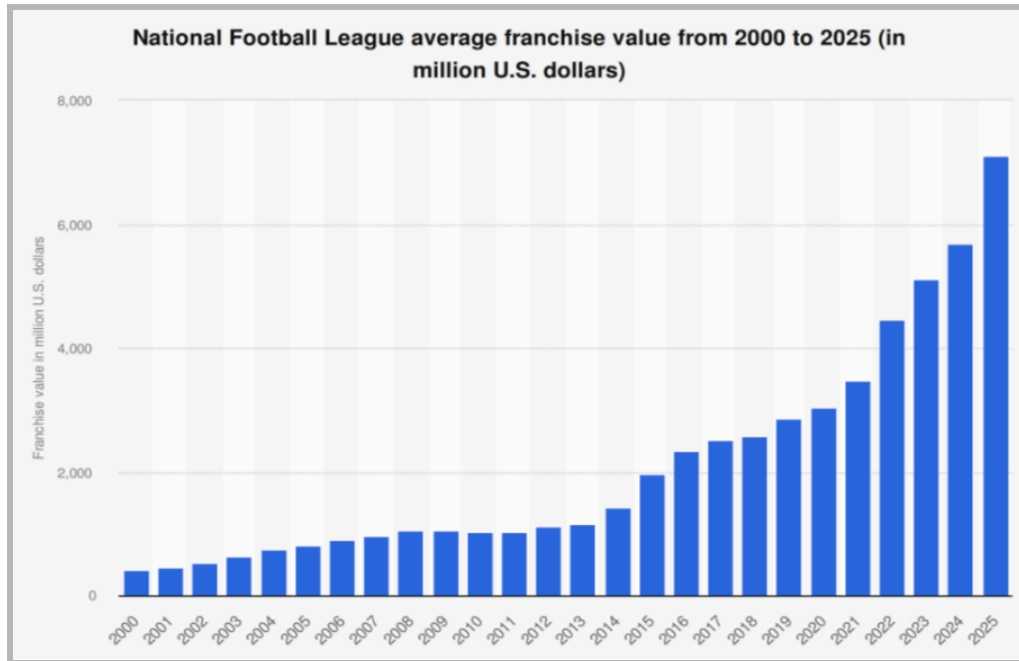


FIGURE 1 *Forbes Average NFL Franchise Valuations, 2000-2025. Statista*

The professional sports industry itself is also growing at a rapid rate, with many of the top sports franchises worldwide increasing in value by nearly 30% year over year. Through the use of Forbes franchise valuations, NFL statistical records, and additional franchise-level data, this study will test whether winning games leads to a statistically significant increase in franchise enterprise value. By using multivariate OLS regression models, variable selection, and a two-way fixed effects model, we test for a positive relationship between on-field success and growth in franchise valuation in the subsequent year.

The results suggest that NFL regular season winning percentage is not statistically significant for predicting franchise value controlling for franchise-specific and league-wide factors. Variables such as fan attendance, superbowl wins, payroll, and stadium characteristics appear to play a larger role in determining NFL franchise value.

INTRODUCTION

In 2024, the *Dallas Cowboys* became the first ever team in the world to surpass a valuation of \$10 billion, and just one year later they were valued at \$13 billion in August of 2025, a 30% increase in just one year. Across the sports world, franchise values have continued to rise at an incredible pace, with multiple organizations seeing valuation increases of over 30% in a single year (Forbes, 2025). Even outside of American sports, Formula One teams like *Scuderia Ferrari* and *Mercedes-AMC Petronas* have seen enormous value increases of over 58%, despite not even being included in the Forbes Top 50 most valuable sports franchises (Forbes, 2025). By using econometric methods to build a model to predict franchise valuation, NFL owners could use such a model to focus on the factors most significant for growing the economic value of a franchise. Elements such as fan base size, league salary cap structures, coaching and management, market size, and the emergence of new athletes every year can all play significant roles in shaping franchise valuation.

RESEARCH

Initially, our goal was to evaluate various sports and leagues to best determine how winning games affects franchise value, however there were numerous challenges in doing so. Every league is structured differently, such as in the total number of games played, player roster sizes,

salary cap regulations, and the total number of teams within each league. It ultimately was difficult to make a balanced comparison between leagues like the MLB and NHL, where MLB teams play double the number of games and operate without a salary cap. After reflection, our research shifted focus towards professional soccer where there was already abundant past econometric literature produced, high global popularity, and many leagues. However, many top soccer leagues use promotion and relegation systems, where the bottom teams are removed from the league each year due to poor performance, making it difficult to create consistent year-over-year analyses. Additionally, there was not enough data accessible and our knowledge of professional soccer was not strong enough.

Ultimately, the NFL is most suitable for the focus of our study for numerous reasons. The league holds a consistent structure over time, maintaining the same 32-team format for the past 24 years. The NFL also operates under the hard salary cap system, ensuring competitive balance where all teams are constricted to spending the same maximum number of dollars on building their roster of players. Therefore, the NFL is a far more controlled environment for analysis compared to alternative pro sports leagues. The NFL provides a substantial amount of accessible public data on franchise valuations, television viewership, fan attendance, stadium characteristics, market size, seating capacity, etc. Most importantly, the league provides clear and consistent records of team success through annual win totals over the last 80 years. And like most sports fans, NFL fans desperately want their teams to win. Recent survey data further supports the importance of winning within professional football, with a 2025 national NFL fan survey identifying team record and on-field success as major factors influencing long-term fan loyalty (QuestionPro, 2025). For many people it is what keeps many fans emotionally invested

and coming back every single season. Ultimately, our research question was narrowed to the following: **How much does winning in the NFL affect the enterprise value of an NFL franchise across different market sizes and seasons?**

METHODOLOGY & DATA

The dataset for this study is a panel dataset with 20 variables and 320 total observations. The 320 observations in the dataset amount from 32 NFL teams over the span of the latest ten-year period (2015-2024 NFL seasons). The twelve most important variables from the data can be studied in Table 1 below:

Variable Name	Variable Type	Description
value	Numeric	Enterprise value of NFL franchise (in billions of USD)
szn_win_pct	Numeric	Team winning % in the regular season
season	Numeric	The NFL season year corresponding to each observation
home_fans	Numeric	Total number of fans in attendance for all home games during the regular season (in thousands)
avg_home_fans	Numeric	Average number of fans attended per home game during the regular season (in thousands)
capacity	Numeric	The total seating capacity inside a team's stadium
stadium_age	Numeric	How many years old a team's stadium is (calculated as: season year - stadium opening year)
new_stadium	Indicator	Indicator variable for if a team is playing their <u>first or second</u> regular season in a new stadium (equal to one if stadium age is less than two years old and equal to zero otherwise)
renovation	Indicator	Indicator variable equal to one if a team's stadium underwent major renovations, equal to zero otherwise
sb_wins	Numeric	The cumulative number of Super Bowl championship won by a

		franchise throughout its history
payroll	Numeric	The total salary cap spent of an NFL franchise on its roster of players in during the regular season (in millions of USD)
dead_cap	Numeric	Total amount of salary cap charge tied up to inactive players no longer on a team's roster (in millions of USD)

TABLE 1 *Variable Descriptions*

The data was collected from numerous sources. Data on the most important variable, team enterprise valuations from the last 10 years, were accessible from the Forbes Business Journal. Team performance data such as games played, wins, and losses for each season were recorded from the NFL's website. Fan attendance and historical super bowl wins data were each collected from ESPN. Payroll data was collected using Spotrac. Data on stadium-related variables such as stadium capacity and opening year were collected using publicly available stadium databases and official NFL team sources. From this information, we constructed variables measuring stadium age, whether a franchise was currently playing in a newly built stadium, and whether a stadium had undergone significant renovations. These variables were included because stadium infrastructure often plays a major role in generating revenue through ticket sales, sponsorships, luxury seating, and fan experience.

The dependent response variable of interest is *value*. ***Value*** measures the total enterprise value of an NFL franchise (in billions of US dollars), reported annually by Forbes in August. Because the NFL regular season happens during the fall each year with the superbowl played in February, valuations in the data are recorded during the year following the corresponding NFL season (ex: the 2016 valuation is recorded for the 2015 NFL season). This is done in order to correctly

estimate the impact of winning in the regular season on franchise value after the season ends. NFL franchises report financials annually in July and Forbes releases valuations in August. Enterprise valuations are likely estimated using financial reports and modeling. The primary explanatory variable of interest is *szn_win_pct*. ***Szn_win_pct*** measures a team's regular season winning percentage. It is calculated as regular season total wins divided by total regular season games played. Therefore, the goal of the study is to estimate the effect of winning percentage on NFL franchise valuation as well as identify other significant predictor variables.

DATA CLEANING

For proper data cleaning, variables *new_stadium* and *renovation* were changed from numeric to categorical (indicator) variables. *New_stadium* is an original created variable indicating if a stadium is new, meaning it is the first or second season it is being used for the team. Therefore, *new_stadium* is equal to one if stadium age is less than two years old and equal to zero otherwise. *Renovation* is another indicator variable equal to one for seasons when new renovations were made to a team's stadium and equal to zero otherwise. These variables are meant to account for any increase to franchise valuation caused by a new stadium or renovations and corresponding attraction of fans that follows. Additionally, variables *home_fans*, *avg_home_fans*, and *capacity* were rescaled to be in thousands. Variables *payroll* and *dead_cap* were rescaled to be in millions of dollars. The primary explanatory variable *szn_win_pct* is rescaled to be put into percentage points for interpretation.

EXPLORATORY DESCRIPTIVE ANALYSES

A correlation matrix, pairwise scatterplots, and histogram of variables were used for conducting exploratory descriptive analyses. According to the correlation matrix, aside from clearly redundant variables, there did not appear to be any significant collinearity between any pairs of predictor variables.

The pairwise scatterplots are split into two groups, team variables and fan/stadium variables for easier visualization and can be seen in the two figures below:

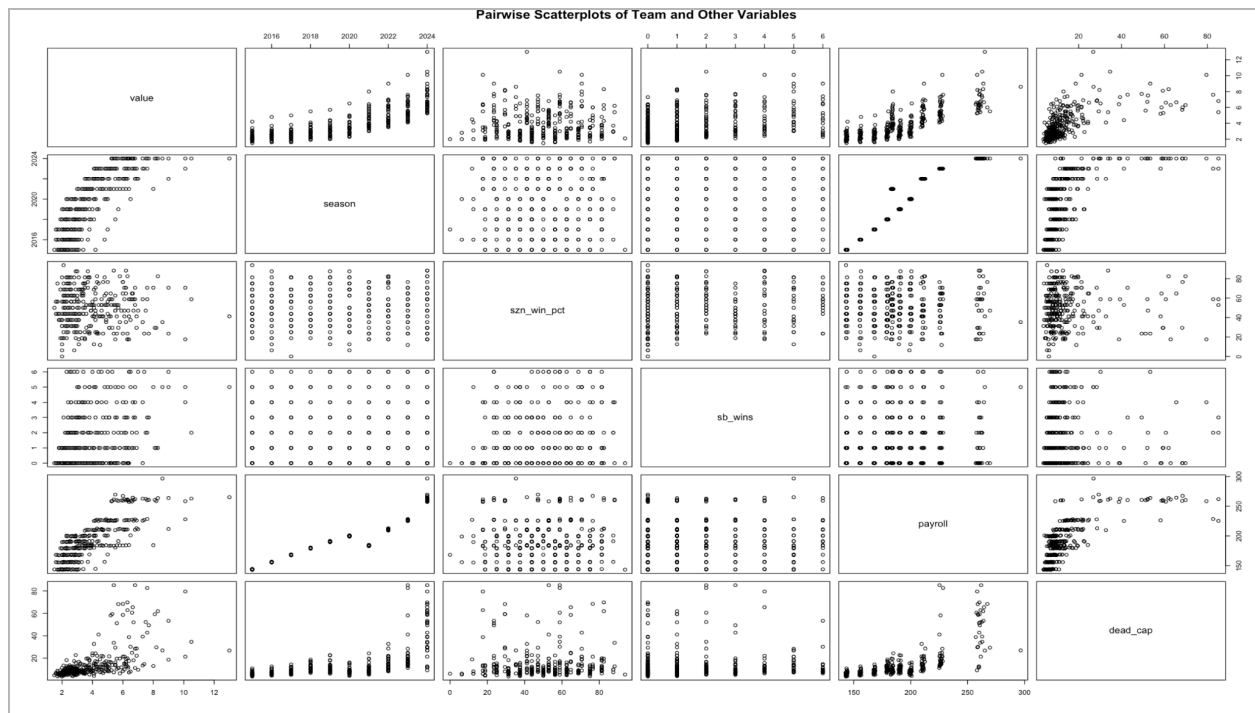


FIGURE 2 *Pairwise Scatterplots for Team Variables: szn_win_pct, sb_wins, payroll, dead_cap*

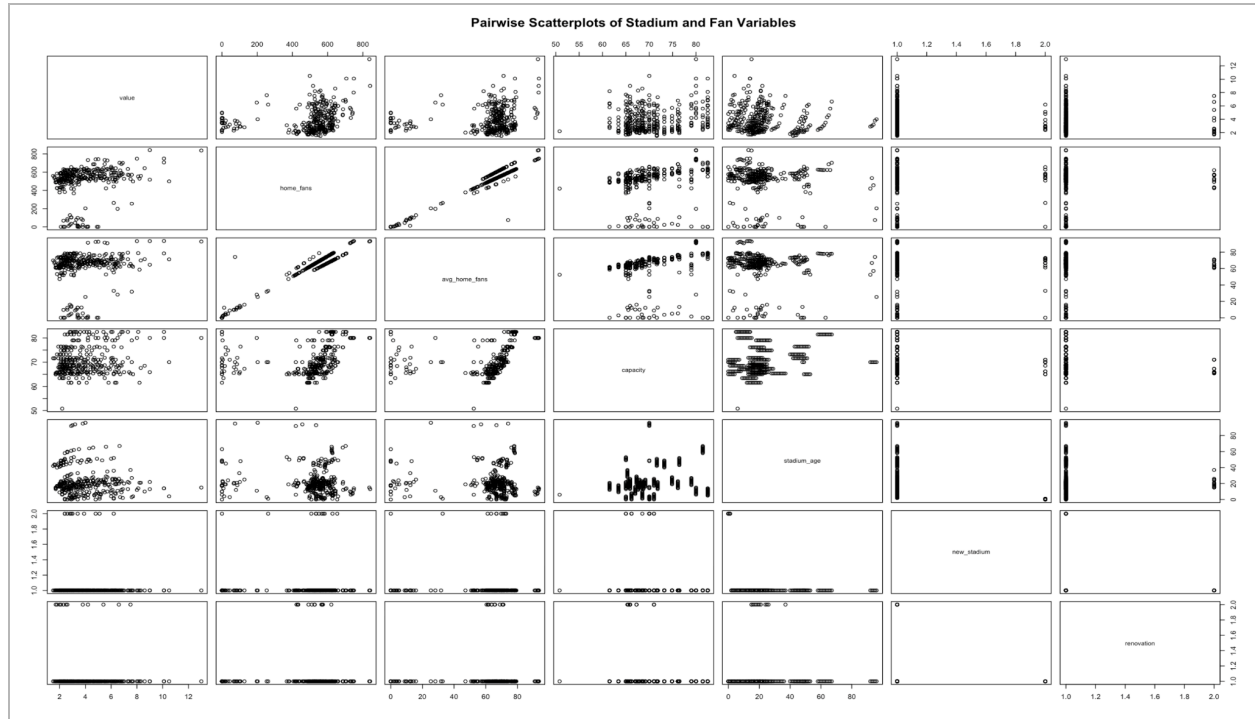


FIGURE 3 *Pairwise Scatterplots for Stadium and Fan Variables*

According to the pairwise scatterplots, there does not appear to be any collinearity between predictors or any need for interaction terms. There does not appear to be any non-linear relationships between any predictors and the response variable, therefore there is no need for higher degree polynomial terms. When building the full regression model, obvious redundant variables are excluded, such as *home_fans* (there is only need to include *avg_home_fans*).

Descriptive summary statistics for the variables in the data can be viewed in Table 2:

	Mean	SD	Min	Max
value	3.93	1.83	1.50	13.00
season	2019.50	2.88	2015.00	2024.00
szn_gp	16.39	0.49	16.00	17.00
szn_wins	8.17	3.13	0.00	15.00
szn_win_pct	49.85	19.09	0.00	93.80
report_date	2020.50	2.88	2016.00	2025.00
home_gp	8.14	0.60	1.00	9.00
home_fans	504.08	175.10	0.00	841.19
avg_home_fans	61.91	20.49	0.00	93.59
capacity	70.57	5.76	50.80	82.50
stadium_open	1997.35	17.21	1923.00	2020.00
stadium_age	22.15	16.97	0.00	96.00
sb_wins	1.67	1.86	0.00	6.00
payroll	192.03	33.27	142.90	296.60
dead_cap	14.75	14.55	3.60	85.21

TABLE 2 *Descriptive Statistics for All Variables*

Additionally, the histograms for the variables can be viewed in Figure 4:

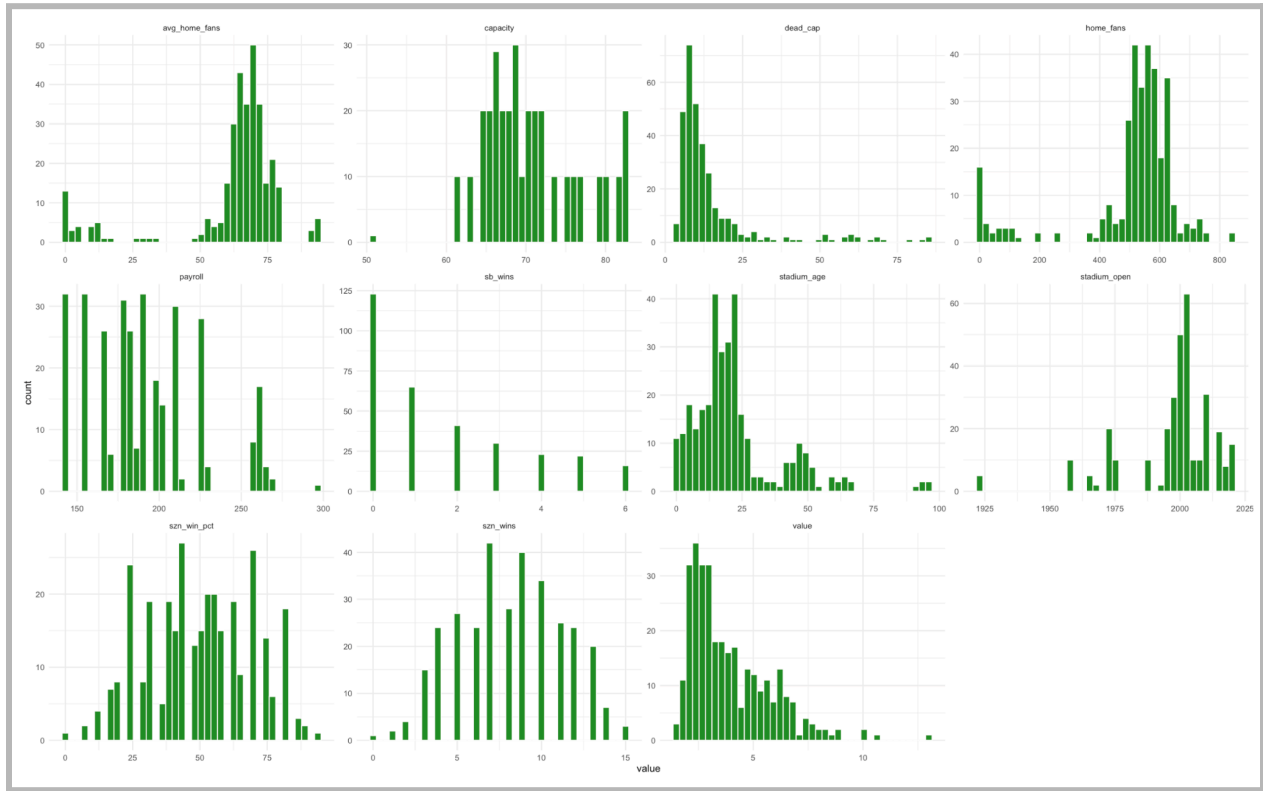


FIGURE 4 *Histograms for All Variables*

MODEL PROCEDURES AND RESULTS

Before fitting any complex regression models with controls, the baseline bivariate OLS regression is fitted using the primary explanatory variable *szn_win_pct* to predict response value.

$$\text{Bivariate OLS : } \widehat{Value} = \beta_0 + \beta_1 szn_win_pct + \mu$$

This model serves as a baseline to improve upon by adding control variables and fixed effects.

After fitting the bivariate OLS model, a full model including all control variables is estimated and can be summarized by the model below:

$$\textbf{Baseline : } \widehat{Value} = \beta_0 + \beta_1 season + \beta_2 szn_win_pct + \beta_3 avg_home_fans + \dots + \beta_{10} sb_wins + \mu$$

A method of backwards selection is then used to remove insignificant variables to reduce model complexity and variance and increase performance in adjusted R^2 . The backwards selection method begins with the fully extended model, then selects the least statistically significant variable with the highest p-value, removing it, then refitting the model and repeating the process until all variables remaining are significant for predicting value. The resulting reduced model is as follows:

$$\textbf{Reduced : } \widehat{Value} = \beta_0 + \beta_1 season + \beta_2 avg_home_fans + \beta_3 capacity + \beta_4 stadium_age + \beta_5 payroll + \beta_6 dead_cap + \beta_7 sb_wins + \mu$$

After using backward selection to select significant variables, they are used to fit a two-way fixed effects model. The two-way fixed effects model uses a team fixed effect to account for any underlying time-invariant differences between team characteristics such as geographic location, market size, and franchise legacy. A second fixed effect for season is also included to control for any time-related shocks impacting all teams in a single season such as the reduced number of fans allowed in stadiums during the 2020 season due to the COVID-19 global pandemic. Cluster-robust standard errors, clustering by team, are used to adjust for any heteroskedasticity and dependence of observations over the numerous time periods. The final two-way fixed effects model is summarized below:

$$\textbf{TWFE Model : } \widehat{value_{it}} = \beta_0 + \beta_1 szn_win_pct_{it} + \beta_2 avg_home_fans_{it} + \beta_3 capacity_{it} + \beta_4 stadium_age_{it} + \beta_5 payroll_{it} + \beta_6 dead_cap_{it} + \beta_7 sb_wins_{it} + \alpha_i + \gamma_t + \mu_{it}$$

The final results of all four models can be observed in Table 3:

NFL Valuation Analysis				
	Bivariate	Baseline (Full)	Reduced Using Backward Selection	TWFE Model
(Intercept)	3.9433*** (0.3597)	-647.9256*** (72.5877)	-650.2270*** (73.6079)	
szn_win_pct	-0.0003 (0.0063)	-0.0024 (0.0030)		0.0005 (0.0014)
season		0.3204*** (0.0366)	0.3215*** (0.0372)	
avg_home_fans		0.0102** (0.0035)	0.0100** (0.0036)	-0.0062*** (0.0015)
capacity		0.0250 (0.0217)	0.0257 (0.0211)	-0.0066 (0.0062)
stadium_age		-0.0136+ (0.0078)	-0.0143* (0.0071)	-0.0127*** (0.0021)
new_stadium1		0.0626 (0.3575)		
renovation1		-0.0124 (0.2707)		
payroll		0.0104* (0.0050)	0.0103* (0.0052)	0.0305** (0.0105)
dead_cap		0.0199** (0.0075)	0.0199** (0.0076)	0.0012 (0.0052)
sb_wins		0.2817*** (0.0802)	0.2781*** (0.0812)	0.1671 (0.1587)
Num.Obs.	320	320	320	320
R2	0.000	0.791	0.791	0.960
R2 Adj.	-0.003	0.785	0.786	0.953
R2 Within				0.143
R2 Within Adj.				0.121
AIC	1300.5	817.0	812.0	359.0
BIC	1311.8	862.2	845.9	539.9
Log.Lik.	-647.267	-396.507	-396.976	
RMSE	1.83	0.84	0.84	0.36
Std.Errors	by: team	by: team	by: team	by: team
FE: team				X
FE: season				X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Standard errors are clustered by NFL Team to account for within-team correlation over time.

TABLE 3 *Results of the Four Models*

According to the results of the four models, it is evident the two-way fixed effects (TWFE) model fits the data best, evident by providing the highest adjusted R^2 value of 0.953, significantly higher than the other three models. TWFE also reports the lowest error metrics for AIC, BIC, and RMSE. TWFE scores an RMSE of 0.36, indicating that its predicted enterprise value is \$360 million away from the observed enterprise value on average, estimating an error that is approximately 3.1% of the full range of the response variable.

By including two-way fixed effects, the results also indicate the coefficient estimate for *szn_win_pct* changes direction from negative to positive, meaning it increases the enterprise value of a franchise, which aligns more with economic intuition. The coefficient estimate of *szn_win_pct* is estimated to be 0.0005, indicating that franchise enterprise value is predicted to increase by \$500,000 on average for every additional percentage point of winning percentage or by \$5 million on average for every additional 10% in winning percentage. However, the p-value indicates a team's season winning percentage is not statistically significant for predicting enterprise value.

Other, more economically relevant, variables such as *avg_home_fans*, *stadium_age*, and *payroll* are statistically significant for predicting enterprise value. *Stadium_age* reports a statistically significant coefficient estimate of -0.0127, indicating that enterprise value is predicted to decrease by approximately \$12.7 million for every additional year a franchise's stadium ages.

Interestingly, *sb_wins*, a significant variable in the preceding three models, becomes insignificant when including fixed effects, possibly because team fixed effects is able to control for some of the "franchise legacy" affect that *sb_wins* would otherwise control for. The TWFE model overall fits the historical data well and can be improved further by removing insignificant variables. An

intriguing extension of the study for future research could be in machine learning by assessing how well the TWFE model can perform on predicting future outcomes using new testing data.

CONCLUSION

This project examined whether winning games meaningfully increases the enterprise value of NFL franchises. We focused on the NFL because it offers a more consistent setting than many other professional sports leagues. All teams operate within the same league structure, play the same number of regular season games, and are subject to the same salary cap system. Using Forbes franchise valuations, NFL team performance data, attendance measures, payroll data, and stadium-level controls, we created a panel dataset of all 32 NFL teams from the 2015 through 2024 seasons, producing 320 total observations. Our main dependent variable was franchise value, measured in billions of dollars, and our main independent variable was regular season winning percentage.

Results suggest that winning alone does not have a statistically significant effect on NFL franchise value when other factors are controlled for. In the simple bivariate model, winning percentage had almost no relationship with franchise value. After adding controls and estimating the TWFE model, the coefficient on winning percentage became positive, which is more intuitive, but it remained statistically insignificant. Therefore, we cannot conclude that better regular season performance leads to higher franchise valuations.

Instead, results provide evidence for other factors, payroll, stadium age, and average home attendance, being more statistically significant in the fixed effects model. Stadium age had a negative relationship with franchise value, suggesting that older stadiums may reduce a team's valuation, likely because newer stadiums create more opportunities for premium seating,

sponsorships, luxury boxes, and improved fan experiences. Payroll was positively related to franchise value, which may reflect the fact that higher-value teams are often able to spend more or are more aggressive in maintaining competitive rosters. Super Bowl wins were significant in earlier models, but became insignificant after team fixed effects were included. This suggests that long-term franchise legacy may already be captured by team-level fixed characteristics rather than by the Super Bowl wins variable alone.

These findings suggest that NFL franchise value is driven by business and structural factors than by short-term winning. While fans often assume winning is the main source of value, a team can still increase in value through strong attendance, a modern stadium, a large market, brand strength, or overall league growth. NFL teams are scarce assets in a highly profitable league, so long-term franchise identity and market position may matter more than one season's record. There are several limitations to this analysis. Forbes valuations are estimates rather than actual precedent sale prices, which may be estimated with numerous assumptions and have measurement errors. The dataset also covers only ten seasons, making it difficult to capture the long-term effects of sustained winning. In addition, the model does not include factors such as media market size, sponsorship revenue, merchandise sales, national exposure, ownership quality, or star players. Winning percentage may also be too broad, since playoff success or multi-year success may better capture the type of winning that affects value.

Overall, this project finds that winning may help a franchise's brand, but it is not the main measurable driver of NFL franchise value in this dataset. Once team and season effects are controlled for, regular season winning percentage is not statistically significant. Future research

could improve the analysis by using more years of data, adding market ad revenue variables, and testing whether playoff success has a stronger effect than regular season winning.